192425037

EXP27:

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

typedef unsigned long long u64;

u64 modpow(u64 b, u64 e, u64 m) {
    u64 r = 1; b %= m;
    while (e) { if (e & 1) r = r * b % m; b = b * b % m; e >>= 1; }
    return r;
}
long long egcd(long long a, long long b, long long *x, long long *y) {
    long long x1, y1, g;
    if (b == 0) { *x = 1; *y = 0; return a; }
    g = egcd(b, a % b, &x1, &y1);
    *x = y1; *y = x1 - (a / b) * y1;
    return g;
}

int main() {
    u64 p = 1009, q = 1013, n = p * q, phi = (p - 1) * (q - 1), e = 65537, d;
    long long x, y;
    u64 table[26], enc[100];
    char msg[100];
    int i, j, len;

    egcd(e, phi, &x, &y);
    d = ((x % (long long)phi) + phi) % phi;
```

```
    printf("Bob's public key: e = %llu, n = %llu\n\n", e, n);

    printf("Alice's message: ");
    scanf("%s", msg);
    len = strlen(msg);
    printf("Ciphertext (each letter A=0..Z=25 encrypted separately):\n  ");
    for (i = 0; i < len; i++) {
        enc[i] = modpow(toupper(msg[i]) - 'A', e, n);
        printf("%llu ", enc[i]);
    }

    /* Attack: only 26 possible plaintext values, so Eve encrypts all of them with
       Bob's PUBLIC key and compares with the intercepted ciphertext numbers. */
    for (i = 0; i < 26; i++) table[i] = modpow(i, e, n);
    printf("\n\nEve's attack (dictionary of 26 encryptions, no factoring of n needed):\n  ");
    for (i = 0; i < len; i++)
        for (j = 0; j < 26; j++)
            if (table[j] == enc[i]) putchar('A' + j);
    printf("\n\nThe method is NOT secure: it is just a monoalphabetic substitution\n");
    printf("cipher with only 26 symbols (frequency analysis or table lookup breaks it).\n");
    return 0;
}
```

OUTPUT:

```
Bob's public key: e = 65537, n = 1022117

Alice's message: HELLO
Ciphertext (each letter A=0..Z=25 encrypted separately):
  292595 162972 20004 20004 586690

Eve's attack (dictionary of 26 encryptions, no factoring of n needed):
  HELLO

The method is NOT secure: it is just a monoalphabetic substitution
cipher with only 26 symbols (frequency analysis or table lookup breaks it).
```